

Module 2: Physical Layer

Study Notes

Contents

1	Data Transmission Concepts	2
2	Guided Media	3
2.1	Twisted Pair Cable	3
2.2	Coaxial Cable	3
2.3	Optical Fiber	3
3	Unguided Media	4
3.1	Radio Waves	4
3.2	VHF (Very High Frequency)	4
3.3	Microwave	4
3.4	Satellite Communication	4
3.5	Infrared	5
4	Encoding Schemes: NRZ, Manchester, Differential Manchester	5
4.1	NRZ (Non-Return to Zero)	5
4.2	Manchester Encoding	5
4.3	Differential Manchester Encoding	5
5	Encoding Scheme Diagrams	5
5.1	NRZ-L Diagram	6
5.2	Manchester Encoding Diagram	6
5.3	Differential Manchester Diagram	6
6	Multiplexing: FDM, TDM, WDM	6
6.1	FDM — Frequency Division Multiplexing	6
6.2	TDM — Time Division Multiplexing	7
6.3	WDM — Wavelength Division Multiplexing	7
7	Circuit Switching, Packet Switching, Message Switching	7
7.1	Circuit Switching	7
7.2	Packet Switching	8
7.3	Message Switching	8

Q1. Data Transmission Concepts

Definition

Data transmission is the process of sending data from one device to another through a communication medium. Several fundamental concepts govern how this works.

1. Analog vs Digital Signals

- **Analog signal:** Continuous waveform that varies smoothly (e.g., sound, AM/FM radio). Susceptible to noise.
- **Digital signal:** Discrete signal with only two states — 0 (low voltage) and 1 (high voltage). More noise-resistant.

2. Simplex, Half-Duplex, Full-Duplex

- **Simplex:** Data flows in one direction only. Example: TV broadcast, keyboard to CPU.
- **Half-Duplex:** Data flows both ways but not simultaneously. Example: Walkie-talkie.
- **Full-Duplex:** Data flows both ways simultaneously. Example: Telephone call.

3. Bandwidth

The range of frequencies a channel can carry, measured in Hz. Higher bandwidth = more data per second.

4. Throughput

The actual amount of data successfully transferred per second (always $\leq$ bandwidth due to overhead/noise).

5. Latency (Propagation Delay)

Time taken for a signal to travel from sender to receiver. Affected by distance and medium speed.

6. Baud Rate vs Bit Rate

- **Baud rate:** Number of signal changes (symbols) per second.
- **Bit rate:** Number of bits transmitted per second. Bit rate = Baud rate $\times$ bits per symbol.

7. Noise and Attenuation

- **Attenuation:** Signal weakens over distance.
- **Noise:** Unwanted electrical signals that distort data. Types: thermal, crosstalk, impulse.

8. Shannon's Capacity Theorem

$$C = B \log_2(1 + \text{SNR})$$

Where C = channel capacity (bps), B = bandwidth (Hz), SNR = signal-to-noise ratio.

Q2. Guided Media

Definition

Guided media (wired) transmits data through a physical conductor. The signal is directed along a specific path. Examples: twisted pair, coaxial cable, optical fiber.

2.1 Twisted Pair Cable

Two insulated copper wires twisted together. Twisting reduces *electromagnetic interference (EMI)* and crosstalk.

Types:

- **UTP (Unshielded Twisted Pair):** No metallic shield. Cheap and easy to install. Used in LAN (Cat5e, Cat6).
- **STP (Shielded Twisted Pair):** Metal foil shield around pairs. Better noise protection. Used in noisy environments.

Specs: Bandwidth up to 250 MHz (Cat6). Distance: up to 100m for Ethernet. Cost: Low.

Use: Telephone lines, office LANs.

2.2 Coaxial Cable

A central conductor surrounded by insulation, a metallic shield (mesh), and an outer plastic jacket. The shield blocks external interference.

Types:

- **Baseband coax:** Digital signal transmission (e.g., older Ethernet 10Base2, 10Base5).
- **Broadband coax:** Analog signal for cable TV (CATV).

Specs: Bandwidth up to 1 GHz. Distance: up to 500m. Cost: Moderate.

Use: Cable TV, older Ethernet networks.

2.3 Optical Fiber

Transmits data as **pulses of light** through a glass or plastic core using *total internal reflection*.

Structure: Core (glass) → Cladding (lower refractive index glass) → Buffer → Jacket.

Types:

- **Single-mode fiber:** Very thin core (8–10 μm). One light path. Long distance, high speed. Used in WANs.
- **Multi-mode fiber:** Wider core (50–62.5 μm). Multiple light paths. Shorter distance. Used in LANs.

Specs: Bandwidth: Tbps range. Distance: up to 100 km (single-mode). Cost: High.

Advantages: Immune to EMI, no electrical hazard, very high speed, lightweight.

Disadvantage: Fragile, expensive to install and splice.

Feature	Twisted Pair	Coaxial	Optical Fiber
Speed	Up to 10 Gbps	Up to 1 Gbps	Tbps
Cost	Lowest	Moderate	Highest
EMI Immunity	Low	Medium	Very High
Distance	100 m	500 m	100 km
Use	LAN, phone	Cable TV	Backbone, ISP

Q3. Unguided Media

Definition

Unguided media (wireless) transmits electromagnetic signals through the air without a physical conductor. Direction is not fixed; signals spread in all directions.

3.1 Radio Waves

Frequency range: 3 KHz – 1 GHz. Omnidirectional (spread in all directions). Can penetrate walls.

- Used in: AM/FM radio, cordless phones, Wi-Fi (2.4 GHz).
- **Pros:** Long range, easy to generate, penetrates obstacles.
- **Cons:** Susceptible to interference, low security.

3.2 VHF (Very High Frequency)

Frequency range: 30 MHz – 300 MHz. Travels in straight lines (line-of-sight).

- Used in: FM radio (88–108 MHz), TV broadcasting (channels 2–13), aviation, marine.
- **Pros:** Good range, relatively clear signals.
- **Cons:** Requires direct line-of-sight for best performance.

3.3 Microwave

Frequency range: 1 GHz – 300 GHz. **Directional** (focused beam). Requires line-of-sight between towers.

- Used in: Mobile phone networks, point-to-point links, radar, satellite links.
- **Pros:** High bandwidth, long distance with relay towers.
- **Cons:** Blocked by buildings, affected by rain (at high frequencies).

3.4 Satellite Communication

A satellite acts as a relay station in orbit. It receives a signal from an earth station (*uplink*), amplifies it, and retransmits it (*downlink*).

Orbit types:

- **GEO** (Geostationary, 36,000 km): Stationary above equator. High latency (270 ms). Used in TV, broadband.
- **MEO** (10,000–20,000 km): Used in GPS. Moderate latency.
- **LEO** (200–2,000 km): Low latency. Used in Starlink, Iridium.

Use: GPS, TV broadcasting, global internet, remote areas.

Cons: High cost, launch complexity, GEO latency.

3.5 Infrared

Frequency: 300 GHz – 400 THz. Very short range, cannot penetrate walls. Line-of-sight required.

- Used in: TV remotes, IrDA (file transfer), proximity sensors.
- **Pros:** No licensing required, no interference with radio.
- **Cons:** Very short range, blocked by obstacles and sunlight.

Q4. Encoding Schemes: NRZ, Manchester, Differential Manchester

Line Encoding

Line encoding converts digital data (bits) into digital signals for transmission. The goal is efficient and error-detectable transmission.

4.1 NRZ (Non-Return to Zero)

The signal does not return to zero between bits.

- **NRZ-L (Level):** Bit value determined by the voltage level. High = 1, Low = 0.
- **NRZ-I (Inverted):** A 1 causes a transition at the start of the bit period; 0 causes no transition.

Problem: Long strings of 0s or 1s cause baseline wander (DC drift) and make it hard to synchronize.

4.2 Manchester Encoding

Each bit period is split in half. A transition **always occurs in the middle** of each bit.

- **1:** Low → High transition in the middle.
- **0:** High → Low transition in the middle.

Used in: Ethernet (10Base-T).

Advantage: Self-clocking — the mid-bit transition provides synchronization automatically.
Disadvantage: Requires double the bandwidth of NRZ.

4.3 Differential Manchester Encoding

Transition at the **start** of the bit period represents a 0; **no transition** at the start represents a 1. There is always a mid-bit transition (for clocking).

- **0:** Transition at the beginning of bit period.
- **1:** No transition at the beginning of bit period.

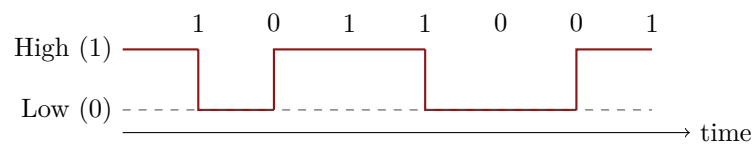
Used in: Token Ring networks.

Advantage: More noise-resistant than Manchester (relies on transition presence/absence, not direction). Still self-clocking.

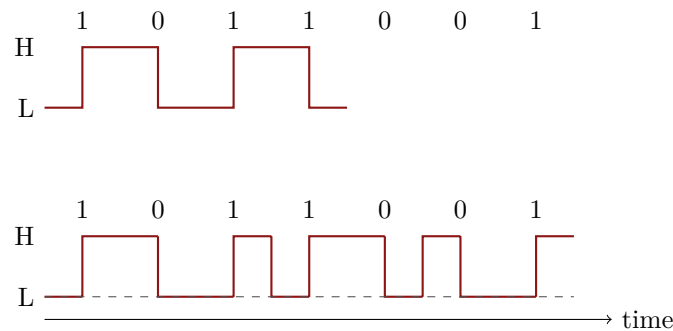
Q5. Encoding Scheme Diagrams

The bit sequence used in diagrams below: **1 0 1 1 0 0 1**

5.1 NRZ-L Diagram

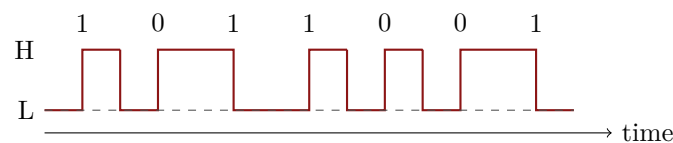


5.2 Manchester Encoding Diagram



Reading the diagram: Each bit occupies one unit of time. The mid-bit transition is the key feature. For Manchester: 1 = rising edge at center; 0 = falling edge at center.

5.3 Differential Manchester Diagram



Key rule: In Differential Manchester, focus on *start-of-bit* transitions: transition present = 0, no transition = 1. Mid-bit transitions are always present (for clock sync only).

Q6. Multiplexing: FDM, TDM, WDM

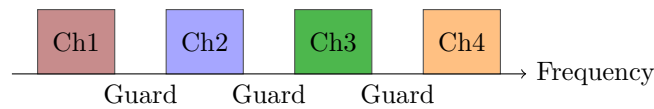
Definition

Multiplexing allows multiple signals to share a single communication channel simultaneously. A device called a **multiplexer (MUX)** combines signals at the sender; a **demultiplexer (DEMUX)** separates them at the receiver.

6.1 FDM — Frequency Division Multiplexing

The available bandwidth of the channel is divided into **non-overlapping frequency bands**. Each sender gets its own frequency band.

- Each signal is modulated onto a different carrier frequency.
- Guard bands separate the channels to prevent overlap.
- **Used in:** AM/FM radio, cable TV, ADSL broadband.

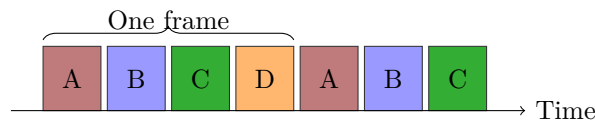


Pros: Simple, works for analog signals, no timing needed.
Cons: Wasteful if some channels are idle; guard bands reduce efficiency.

6.2 TDM — Time Division Multiplexing

The channel is divided into **time slots**. Each sender gets the full bandwidth but only for its assigned time slot.

- **Synchronous TDM:** Fixed slots assigned to each user, even if they have no data. Simple but wasteful.
- **Statistical (Asynchronous) TDM:** Slots assigned dynamically only when a user has data. More efficient.
- **Used in:** T1/E1 telephone lines, SONET, digital telephony.



Pros: Full bandwidth per slot; suitable for digital signals; no interference.
Cons: Synchronous TDM wastes slots when idle.

6.3 WDM — Wavelength Division Multiplexing

Multiple **light wavelengths (colors)** are transmitted simultaneously over a single optical fiber. Conceptually similar to FDM but for optical signals.

- **DWDM (Dense WDM):** Packs many wavelengths closely (up to 160+ channels). Used in long-haul fiber networks.
- **CWDM (Coarse WDM):** Fewer channels, wider spacing. Cheaper, shorter distances.
- **Used in:** Internet backbone, fiber-optic ISP networks.

Pros: Massively increases fiber capacity; each wavelength can carry Gbps.
Cons: Expensive equipment (lasers, prisms); only works with optical fiber.

Feature	FDM	TDM	WDM
Division basis	Frequency	Time	Wavelength (light)
Signal type	Analog	Digital	Optical
Medium	Copper/Air	Any	Optical fiber only
Example	FM radio, ADSL	T1 lines	Internet backbone

Q7. Circuit Switching, Packet Switching, Message Switching

7.1 Circuit Switching

A **dedicated physical path** is established between sender and receiver before communication begins and is maintained for the entire session.

Phases: (1) Circuit establishment → (2) Data transfer → (3) Circuit teardown.

Example: Traditional telephone (PSTN).

Pros: Guaranteed bandwidth; no delay once circuit is set up; suitable for real-time voice.

Cons: Wastes bandwidth when idle; long setup time; circuit fails if any link breaks.

7.2 Packet Switching

Data is broken into small units called **packets**. Each packet travels independently through the network, possibly via different routes, and is reassembled at the destination.

Two approaches:

- **Datagram (connectionless):** Each packet routed independently. Used in IP networks.
- **Virtual circuit (connection-oriented):** Path decided in advance; all packets follow same route (e.g., ATM).

Example: Internet (IP), Wi-Fi.

Pros: Efficient bandwidth use; fault-tolerant (rerouting); supports variable traffic.

Cons: Variable delay (jitter); packets may arrive out of order.

7.3 Message Switching

The entire message is stored at each intermediate node, then forwarded to the next node when a link is free. Also called **store-and-forward**.

Example: Email, early telegraph systems.

Pros: No dedicated circuit needed; messages can be stored if destination is busy.

Cons: High delay; requires large storage at intermediate nodes; not suitable for real-time.

Feature	Circuit Switching	Packet Switching	Message Switching
Path	Dedicated fixed	Dynamic per packet	Dynamic per message
Delay	Low (after setup)	Low-moderate	High
Bandwidth	Reserved/wasted	Shared efficiently	Shared
Store & Forward	No	Per packet	Whole message
Failure handling	Circuit breaks	Reroute packets	Store at node
Example	PSTN (telephone)	Internet (IP)	Email, SMS
Real-time use	Yes	Yes (with QoS)	No